

Dobot Magician Lite Pick-and-Place Operations Using Suction Cup and Gripper End Effectors

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ABSTRACT

This report presents hands-on experiments with the Dobot Magician Lite across two lab exercises involving a suction cup and a gripper. Using waypoint teaching and Python integration through the pydobot, repeatable pick-and-place routines were developed and tested. Key procedures included end-effector setup, pallet geometry definition, and cycle-time evaluation. Challenges such as intermittent communication, vacuum seal loss, and gripper misalignment were mitigated through parameter tuning and system checks. Results demonstrated trade-offs between reliability and efficiency, while underscoring the platform's instructional value and potential for advanced robotics and automation applications.

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1. INTRODUCTION

Robotic arms play a central role in modern automation, enabling precise, repeatable movements across industries such as manufacturing, packaging, and material handling. The Dobot Magician Lite provides a safe, educational platform to study these robotic actions through interchangeable end-effectors.

The innovation lies in integrating waypoint teaching with Python-based control, allowing students to explore motion planning, end-effector configuration, and repeatable code-driven routines. Challenges such as intermittent communication, vacuum seal issues, and alignment drift highlighted trade-offs in reliability, while reinforcing programming skills and system troubleshooting. Overall, the lab emphasizes both the practical and instructional value of the Dobot Magician Lite in robotics and automation.

❖ Problem Statement:

The problem addressed in this lab is how suction cups and grippers differ in cycle time, accuracy, and handling consistency. The assignment required performing pallet-to-pallet pick-and-place tasks with both effectors, recording outcomes, and comparing which tool is better suited for different object types.

2. METHOD

❖ Experimental Setup:

The experimental platform was based on the Dobot Magician Lite, a desktop robotic manipulator designed with an interchangeable end-effector interface. The study simulated a scaled industrial pallet transfer under controlled laboratory conditions. Two effectors were evaluated: a compliant suction cup for smooth lightweight parts and a parallel-jaw gripper for irregular objects.

The robot was operated via an Ubuntu 22.04 LTS workstation to ensure long-term software compatibility, with communication established through Python 3.10 and the pydobot2 library. This framework provided high-level motion primitives (e.g., `move_to`) and effector commands (`.suck()`, `.grip()`). A total of ten target waypoints were defined, including a home pose, an intermediate clearance position, and eight pallet corners.

Palletized plastic cubes served as the workpieces, with four units transferred sequentially from the source to the destination pallet. This setup enabled a systematic comparison of effector performance in terms of reliability, speed, and handling precision.

2.1 Using Suction Cup End Effector:

The suction cup effector was mounted on the Dobot Magician Lite to evaluate its performance in pallet-to-pallet transfers. At the beginning of each trial, the manipulator was initialized by powering on the system, establishing communication through the verified COM port, and executing the `.home()` command to define a consistent reference frame.

Operational waypoints were then recorded. The robot was manually jogged to the corners of the source and destination pallets, and positional data were extracted using the `get_pose()` function. These coordinates, expressed in terms of X, Y, Z, and R values, were stored in the Python control program. The Z-coordinate of each point was subsequently adjusted to ensure reliable contact with the cubes while avoiding collision with the workspace.

Program execution relied on Python scripts implemented through the `pydobot2` interface. Motion commands were issued via the `device.move_to()` function in `MOVJ_XYZ` mode, while suction was actuated using `device.suck(True)` for engagement and `device.suck(False)` for release. A delay interval of one to two seconds was inserted between these actions to accommodate the vacuum system's response time.

The task sequence began with the robot moving from the home pose to an intermediate clearance point, followed by transitions to the designated source pallet corners. At each source location, suction was engaged to grasp a cube, after which the robot transported the object through the clearance waypoint and released it at the corresponding destination corner. To be noted that, this cycle was repeated until four cubes had been transferred.

Cycle time was quantified using Python's time library by recording the elapsed duration from the initial pickup to the final placement. This procedure facilitated systematic evaluation of the suction cup's efficiency, repeatability, and sensitivity to Z-axis alignment in multiple experimental runs.

2.2 Using Gripper End Effector:

Following completion of the suction trials, the mechanical gripper was installed in place of the suction cup to assess its comparative performance. The robot was reinitialized and homed to ensure accurate calibration after the effector exchange. For consistency, the same home, intermediate, and pallet corner waypoints defined in the suction experiments were reused, enabling direct comparison between the two tools.

The control program was adapted to reflect gripper operation. Actuation commands were executed through `device.grip(True)` to close the jaws and `device.grip(False)` to release them, while motion was performed using the same waypoint sequence established earlier. To accommodate the mechanical response of the gripper, additional delays of approximately two seconds were inserted at both pickup and placement locations.

The operational cycle began at the home pose and proceeded through the intermediate clearance point before approaching each source pallet position. At each source corner, the gripper was closed to secure a cube, lifted through the clearance waypoint, and opened at the corresponding destination corner to complete transfer. This sequence was repeated until four cubes were relocated.

Cycle time was quantified using Python's time library by recording the elapsed duration from the initial pickup to the final placement. The procedure enabled systematic evaluation of the gripper's adaptability to object geometry, positional accuracy, and durability relative to the suction cup trials.

3 RESULTS AND DISCUSSION

The outcomes of the pick-and-place experiments using suction and gripper end effectors, followed by a comparative discussion is shown in this section. Observations were made in terms of precision, cycle time, reliability, and consistency.

3.1 Suction Cup

Trials with the suction cup effector demonstrated strong performance when interacting with smooth plastic cubes. When the Z-axis height was correctly adjusted, the suction reliably lifted the objects without slippage. However, improper calibration introduced notable failure modes: excessive clearance resulted in incomplete vacuum sealing, whereas overextension caused collisions with the pallet surface.

In terms of cycle time, the suction cup achieved faster transfers than the gripper. This improvement was attributed to the immediate engagement and release of the vacuum system compared to the mechanical la

of the gripper jaws. Nevertheless, reliability was object-dependent. While the suction cup performed well with smooth and uniform cubes, it struggled with porous, irregular, or heavier objects.

Failures occurred in the form of incomplete lifts or mid-transfer drops. Consistency was high when objects were identical, but performance degraded significantly with variation in surface texture or geometry. These results indicate that suction excels in speed but is limited in versatility and robustness. Figure 3.1 shows the Dobot Magician Lite equipped with a suction cup effector is lifting a cube from the pallet workspace.



Figure 3.1. Suction cup effector lifting a cube from the pallet

3.2 Gripper

Experiments with the parallel-jaw gripper highlighted different trade-offs. Precision was contingent on accurate XY alignment: misaligned approaches often displaced the cube rather than securing it. Once aligned, however, the grip was stable and repeatable.

Cycle times were longer than those of the suction cup, primarily due to two factors: the additional seconds required for mechanical opening and closing, and cautious approach motions to avoid misplacement. Despite this penalty, the gripper consistently demonstrated greater reliability. It successfully handled all test cubes regardless of surface finish, minor irregularities, or weight differences.

No failures such as dropped objects were observed across repeated trials. Consistency was nearly perfect, though dependent on operator care in alignment. Figure 3.2 shows the Dobot Magician Lite equipped with a mechanical gripper effector is lifting a cube from the pallet workspace.



Figure 3.2. Gripper effector lifting a cube from the pallet

3.3 Comparative Analysis of Suction Cup and Gripper End Effectors:

Overall, the suction cup proved advantageous in speed but was highly sensitive to object material and surface properties. In contrast, the gripper was slower but robust, maintaining reliability across a wider range of conditions.

These findings highlight the classic trade-off between efficiency and versatility in robotic end-effector design. For industrial and educational settings, suction is preferable for uniform, lightweight components, while the gripper is better suited for irregular or variable objects where reliability is paramount. A systematic comparison of the two effectors is provided in Table 1.

Parameter	Suction Cup	Gripper
Precision	Reliable when Z-axis height is correctly tuned; errors arise from improper vacuum sealing or collisions.	Requires accurate XY alignment; once aligned, grip is stable and repeatable.
Cycle Time	Faster due to immediate engagement and release of vacuum system.	Slower due to mechanical opening/closing and cautious approach motions.
Reliability	High for smooth, lightweight cubes; prone to failure with porous, irregular, or heavy objects.	Consistently successful across all tested objects, irrespective of surface finish or minor irregularities.
Consistency	Stable for identical objects; performance declines with variations in texture or geometry.	Nearly 100% success across repeated trials, contingent on proper alignment.
Best Suited For	High-speed transfers of uniform, lightweight parts.	Robust handling of diverse, irregular, or heavier objects.

Table 1. Suction vs. Gripper Performance

❖ DISCUSSIONS:

- Before this experiment, fundamental principles of robotic waypoint teaching and end-effector actuation were known conceptually but not practiced in depth. In this lab, approximately ten points were defined, including the home position, an intermediate clearance point, and the eight pallet corners, which provided sufficient spatial coverage for repeatable pallet-to-pallet transfers.
- The Python implementation required a modest number of lines, primarily for initializing communication, defining taught points, and sequencing motion commands. Initial scripts were longer due to repetitive function calls, but efficiency can be improved by modularizing the code, defining reusable functions for pickup and placement, and incorporating loop structures to automate repeated cube transfers.
- The gripper was observed to be more reliable than the suction cup. While suction achieved faster cycle times under ideal conditions, it frequently failed with irregular or porous surfaces. By contrast, the gripper consistently succeeded across all trials, independent of cube geometry or weight variations, making it more suitable for industrial settings where variability is common.
- Object characteristics had a decisive effect on performance. The suction cup was highly effective with smooth, lightweight objects but unreliable with porous or textured materials, where the vacuum seal could not be maintained. The gripper exhibited greater adaptability, handling variations in surface finish and shape without loss of performance, which increases its industrial applicability.
- In practice, minor adjustments to the Z-axis values of the taught points were required to ensure proper contact with the cubes while avoiding collisions with the pallet surface. Once calibrated, these points required little modification between runs, confirming the repeatability of the system.
- The complete implementation required several dozen lines of code, depending on the level of detail in motion and delay instructions. Efficiency was improved by reducing redundancy through the use of loops, parameterized functions, and structured programming practices. This approach not only minimized the codebase but also enhanced readability and scalability for more complex tasks.

4 CONCLUSION

This lab involved pallet-to-pallet pick-and-place actions utilizing the Dobot Magician Lite with suction and gripper effectors. The two approaches were written in Python utilizing the pydobot2 library and did the job successfully.

The suction control is governed by the `device.suck(True)` and `device.suck(False)` commands, which must be synchronized with motion instructions. To ensure proper pickup, the suction is activated immediately after the end effector reaches the target Z-coordinate and before any lifting motion is executed.

A short-programmed delay of one to two seconds following activation allows the vacuum system to stabilize and secure the object. Similarly, reliable release is obtained by deactivating suction once the end effector reaches the destination pallet, again with a short delay to allow the part to disengage completely before retracting. This temporal adjustment ensures that suction is neither activated prematurely nor disengaged too late, thereby improving task accuracy.

When objects other than the plastic cubes were introduced, the suction system displayed variable performance depending on surface characteristics and material properties. Smooth, non-porous objects of comparable weight were successfully manipulated, as the vacuum seal could be maintained. In contrast, porous, irregularly shaped, or heavier items led to frequent failures, including incomplete pickup, loss of grip during transfer, or total inability to lift the object. These observations confirm that suction is highly dependent on surface finish and object geometry, and its reliability diminishes significantly when operating on items that deviate from ideal test conditions.

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REFERENCES

- [1] RAS 545 Laboratory Manual – Lab 2: Dobot Magician Lite Pick & Place
- [2] *pydobot2* Documentation. Available: <https://pypi.org/project/pydobot2/>

APPENDIX

- Python Scripts:
 - suction_robotics_J1.py

```
• import pydobot
• import time
•
• # Connect to the Dobot
• device = pydobot.Dobot(port="COM9")
•
• # Make sure suction is OFF at the start (nothing held)
• device.suck(False)
• time.sleep(1)
•
• # Safety height to move up/down (z)
• safe_z = 15
• place_z = -45
•
• # Positions format: [x, y, z, r]
• palA_positions = [
•     [251.00, -12.60, -44.55, 0], # 1
•     [251.57, -71.16, -42.93, 0], # 2
•     [311.00, -12.06, -42.26, 0], # 3
•     [311.05, -71.59, -42.32, 0] # 4
• ]
•
• palB_positions = [
•     [251.00, 64.00, -43.50, 0], # 1
•     [251.55, 9.33, -42.93, 0], # 2
•     [312.50, 65.00, -41.50, 0], # 3
•     [311.05, 7.66, -42.72, 0] # 4
```

```

• ]
•
• # ----- Helper function -----
• def pick_and_place(pick, place):
•     px, py, pz, pr = pick
•     dx, dy, dz, dr = place
•
•     # ---- Pick ----
•     device.move_to(x=px, y=py, z=safe_z, r=pr)
•     device.move_to(x=px, y=py, z=place_z, r=pr)
•     device.suck(True)      # suction ON (pick)
•     time.sleep(1)
•     device.move_to(x=px, y=py, z=safe_z, r=pr)
•
•     # ---- Place ----
•     device.move_to(x=dx, y=dy, z=safe_z, r=dr)
•     time.sleep(0.5)
•     device.move_to(x=dx, y=dy, z=place_z, r=dr)
•     device.suck(False)    # suction OFF (release)
•     time.sleep(1)
•     device.move_to(x=dx, y=dy, z=safe_z, r=dr)
•
• # ----- Main Loop -----
• # Forward trip: A -> B
• for i in range(len(palA_positions)):
•     pick_and_place(palA_positions[i], palB_positions[i])
•
• # Return trip: B -> A (reverse order so no block is skipped)
• for i in reversed(range(len(palB_positions))):
•     pick_and_place(palB_positions[i], palA_positions[i])
•
• # ◇ Ensure suction is OFF after the last block
• device.suck(False)
• time.sleep(1)
•
• print("☑ Task completed: all blocks moved A→B and back B→A with
• suction tool.")
•

```

○ gripper_robotics_J1.py

```

• import pydobot
• import time
•
• # Connect to the Dobot
• device = pydobot.Dobot(port="COM9")
• device.speed (10,10)

```

```

• # Open gripper at the start (make sure nothing is held)
• device.grip(False)
• time.sleep(1)
•
• # Safety height to move up/down (z)
• safe_z = 0 # don't change
• place_z = -22
•
• # Positions format: [x, y, z, r]
• palA_positions = [
•     [251.00, -12.60, 0, 0], # 1
•     [251.57, -71.16, 0, 0], # 2
•     [311.00, -12.06, 0, 0], # 3
•     [311.05, -71.59, 0, 0] # 4
• ]
•
• palB_positions = [
•     [251.00, 67.00, 0, 0], # 1
•     [251.55, 9.33, 0, 0], # 2
•     [312.50, 65.00, 0, 0], # 3
•     [311.05, 7.66, 0, 0] # 4
• ]
•
• # ----- Helper function -----
• def pick_and_place(pick, place):
•     px, py, pz, pr = pick
•     dx, dy, dz, dr = place
•
•     # ---- Pick ----
•     device.move_to(x=px, y=py, z=safe_z, r=pr)
•     time.sleep(1)
•     device.move_to(x=px, y=py, z=place_z, r=pr)
•     device.grip(True) # Close gripper
•     time.sleep(1)
•     device.move_to(x=px, y=py, z=safe_z, r=pr)
•     time.sleep(1)
•
•     # ---- Place ----
•     device.move_to(x=dx, y=dy, z=safe_z, r=dr)
•     time.sleep(1)
•     device.move_to(x=dx, y=dy, z=place_z, r=dr)
•     device.grip(False) # Open gripper
•     time.sleep(1)
•     device.move_to(x=dx, y=dy, z=safe_z, r=dr)
•     time.sleep(1)
•
• # ----- Main Loop -----

```

```

• # Forward trip: A -> B
• for i in range(len(palA_positions)):
•     pick_and_place(palA_positions[i], palB_positions[i])
•
• # Return trip: B -> A (reverse order so no block is skipped)
• for i in reversed(range(len(palB_positions))):
•     pick_and_place(palB_positions[i], palA_positions[i])
•
• # ◇ Ensure gripper is open after the last block
• device.grip(False)
• time.sleep(1)
•
• print("☑ Task completed: all blocks moved A→B and back B→A, gripper
  open.")

```

- Video Demonstration:
 - Gripper: <https://youtu.be/cRmamC6WOwo>
 - Suction: <https://youtu.be/4O1vtqpEcjQ>
 - AI Usage: Generative AI (ChatGPT) was used for debugging Python code, improving timing logic, and preparing this report. The full chat link is included in the submission for transparency
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